Chapter 9: Emotion and Motivation Mathematical Equations

Comprehensive Neuroscience Reference

Abstract

This document presents the mathematical framework underlying emotion and motivation neuroscience. We develop equations for appraisal theories (multi-dimensional emotional scoring), circumplex models of emotion (valence-arousal space), reward learning via temporal difference algorithms and dopamine, fear conditioning and extinction, prefrontal regulation of amygdala responses, interoceptive body state representation, HPA axis stress response dynamics with cortisol kinetics, heart rate variability and vagal tone assessment, and social bonding through neurochemical coupling. Six detailed case studies with numerical simulations ground the theory in quantitative predictions relating to behavioral, physiological, and neuroimaging data.

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1. Appraisal Theories of Emotion

1.1 Multi-Dimensional Appraisal Scoring

Appraisal theories (Lazarus, Scherer) propose that emotions arise from evaluations of events along multiple cognitive dimensions:

$$A_{\text{total}} = w_1 G_r + w_2 G_c + w_3 A_g + w_4 C_p + w_5 C_a \tag{1}$$

Symbol	Meaning	Scale
A_{total}	Overall appraisal intensity (emotional intensity)	0–10
G_r	Goal relevance: personal goal involvement	0–1
G_c	Goal congruence: outcome-goal alignment	-1 to $+1$
A_g	Agency: internal vs. external responsibility	-1 to $+1$
C_p	Coping potential: manageability	0–1
C_a	Coping actual: success of coping	0–1
w_i	Weights ($\sum w_i = 1$)	Typical: 0.25, 0.25, 0.15, 0.2, 0.15

A stimulus that is highly relevant to goals, incongruent with those goals, externally caused, and uncontrollable produces a strong emotional response.

1.2 Emotion Intensity from Appraisal Dimensions

$$I_{\text{emotion}} = G_r \times |G_c| \times (1 - C_p) \tag{2}$$

Emotional intensity is maximized when an event is personally relevant (G_r high), inconsistent with goals ($|G_c|$ large), and uncontrollable (low C_p).

2. Circumplex Model of Emotion

2.1 Emotion Position in 2D Valence-Arousal Space

The circumplex model represents emotions as positions in 2D space spanned by **valence** (positive vs. negative) and **arousal** (high vs. low):

$$V = V_{\max} \cos(\theta - \theta_0) \tag{3}$$

$$A = A_{\text{max}} + A_{\text{baseline}} \sin(\theta - \theta_0) \tag{4}$$

Symbol	Meaning	Range
V	Valence	-1 to $+1$
A	Arousal	0–1
θ	Angle in emotion space	0°–360°
θ_0	Reference angle (emotion category)	Emotion-dependent
V_{max}	Valence amplitude	1.0
A_{max}	Arousal amplitude	0.5

Each basic emotion occupies a characteristic region: joy (positive, high arousal), sadness (negative, low arousal), fear (negative, high arousal), contentment (positive, low arousal).

2.2 Emotional Intensity in Circumplex Space

$$I_{\text{emotion}} = \sqrt{(V_{\text{dev}})^2 + (A_{\text{dev}})^2} \tag{5}$$

The intensity is the Euclidean distance from the neutral point (V=0, A=baseline). Strong emotions are far from the center; weak emotions are near the center.

3. Reward and Motivation: Temporal Difference Learning

3.1 Dopamine Reward Prediction Error

Dopamine neurons in VTA and SNc encode the temporal difference error — the discrepancy between expected and actual reward:

$$\delta(t) = R(t) + \gamma V(s_{t+1}) - V(s_t) \tag{6}$$

$$\Delta V(s_t) = \alpha\, \delta(t) \tag{7}$$

Symbol	Meaning	Typical Value
$\delta(t)$	Reward prediction error (RPE)	Arbitrary units
$R(t)$	Actual reward at time t	Scalar
$V(s_t), V(s_{t+1})$	Value of current and next state	

Symbol	Meaning	Typical Value
γ	Discount factor	0.9–0.99
α	Learning rate	0.01–0.5

When $\delta > 0$ (reward > expectation), dopamine fires, strengthening synapses. When $\delta < 0$ (reward < expectation), dopamine pauses, weakening synapses.

3.2 Dopamine Firing Rate as Function of Prediction Error

$$\text{Firing rate} = f_{\text{baseline}} + k\,[R(t) - V(s_t)] \tag{8}$$

Symbol	Meaning	Value
Firing rate	Spike frequency of VTA/SNc dopamine neurons	spikes/s
f_{baseline}	Baseline firing in absence of reward signal	2–5 spikes/s
k	Sensitivity (gain) to prediction error	0.1–0.5 spikes/s per unit

Dopamine neurons show a near-linear response to reward prediction error over a wide range.

4. Fear Conditioning and Extinction

4.1 Amygdala Fear Learning

Fear conditioning follows a simple Pavlovian learning rule:

$$\Delta W_{\text{CS}\to\text{CeA}} = \beta_{\text{acq}} \times \lambda_{\text{US}} \times (1 - W_{\text{CS}\to\text{CeA}}) \tag{9}$$

where:

- $W_{\text{CS}\to\text{CeA}}$ = synaptic weight from conditioned stimulus (tone) to central amygdala (fear output)
- β_{acq} = acquisition learning rate ($\sim$0.1–0.3)
- λ_{US} = unconditioned stimulus (shock) presence (1 if shock present, 0 if absent)

After pairing, W increases toward 1. The CS alone then activates CeA, producing fear responses (freezing, potentiated startle, autonomic arousal).

4.2 Extinction Learning via Vmedia PFC

During extinction (repeated CS without US), vmPFC learns to suppress CeA via intercalated cell masses:

$$\frac{d}{dt}\text{VmediaPFC}_{\text{inhibition}} = \alpha_{\text{ext}} \times (1 - \text{VmediaPFC}_{\text{inhibition}}) \tag{10}$$

$$\text{Fear output} = W_{\text{CS}\to\text{CeA}} \times (1 - \text{VmediaPFC}_{\text{inhibition}}) \tag{11}$$

With repeated extinction trials, vmPFC inhibition grows, suppressing the fear response. However, extinction is slower than acquisition (slower α_{ext} than β_{acq}), explaining why exposure therapy requires many trials and why spontaneous recovery of fear occurs.

5. Prefrontal Regulation of Emotion

5.1 Cognitive Reappraisal via dlPFC

Dorsolateral PFC (dlPFC) can reappraise emotional stimuli, modulating amygdala response:

$$\text{Amygdala response} = (\text{Stimulus valence}) \times (1 - \text{Reappraisal factor}) \tag{12}$$

$$\text{Reappraisal factor} = \frac{\text{dlPFC activity}}{1 + e^{-\beta(\text{dlPFC activity} - \theta)}} \tag{13}$$

where β determines steepness of the suppression curve, and θ is the dlPFC activity threshold. Strong reappraisal (high dlPFC activity) suppresses amygdala; weak reappraisal allows amygdala to respond fully.

6. Interoception and Embodied Emotion

6.1 Body State Representation in Anterior Insula

The anterior insula generates a real-time representation of visceral body state (interoception):

$$\text{Interoceptive state} = \sum_i w_i \cdot I_i(t) \tag{14}$$

where $I_i(t)$ are interoceptive signals (heart rate, respiration, skin conductance, gut distension, temperature, etc.) and w_i are integration weights.

The insula's subjective feeling corresponds to this integrated body state coupled with emotional context (amygdala valence + PFC appraisal).

7. Stress Response: HPA Axis Dynamics

7.1 CRH and ACTH Dynamics

The hypothalamic-pituitary-adrenal axis mediates sustained stress response:

$$\frac{d[\text{CRH}]}{dt} = k_{\text{stim}} \cdot \text{Stress} - k_{\text{inhib}} \cdot [\text{Cortisol}] - k_{\text{deg}} \cdot [\text{CRH}] \tag{15}$$

$$\frac{d[\text{ACTH}]}{dt} = \beta_{\text{CRH}} \cdot [\text{CRH}] - k_{\text{deg,ACTH}} \cdot [\text{ACTH}] \tag{16}$$

$$\frac{d[\text{Cortisol}]}{dt} = \beta_{\text{ACTH}} \cdot [\text{ACTH}] - k_{\text{clearance}} \cdot [\text{Cortisol}] \tag{17}$$

Key parameters:

- Stress stimulus from amygdala and brainstem
- Cortisol provides negative feedback (inhibits CRH and ACTH)
- Chronic stress impairs negative feedback → sustained HPA activation

7.2 Cortisol Kinetics and Hippocampal Effects

Cortisol binds to glucocorticoid receptors (GR) in hippocampus with concentration-dependent effects:

$$\text{GR occupancy} = \frac{[\text{Cortisol}]}{K_d + [\text{Cortisol}]} \tag{18}$$

$$\text{Hippocampal effect} = (\text{GR occupancy}) \times (\text{Maximal effect}) \times (1 - \text{Chronic adaptation}) \tag{19}$$

Acute moderate cortisol (∼1–2 h) enhances memory; chronic elevated cortisol (> weeks) causes dendritic retraction, reduced neurogenesis, volume reduction.

8. Heart Rate Variability and Vagal Tone

8.1 RR Interval Variability

Heart rate variability (HRV) reflects parasympathetic (vagal) modulation of cardiac rate:

$$\text{RMSSD} = \sqrt{\frac{1}{N-1} \sum_{i=1}^{N-1} (\text{RR}_i - \text{RR}_{i+1})^2} \tag{20}$$

where RR intervals are successive heartbeat intervals (in milliseconds).

- High RMSSD (>50 ms): Excellent parasympathetic tone; good stress regulation
- Moderate (25–50 ms): Normal; adequate emotional flexibility
- Low (10–25 ms): Reduced parasympathetic tone; stress or anxiety
- Very low (<10 ms): Poor vagal tone; autonomic dysregulation

8.2 Heart Rate Variability Frequency Bands

The power spectral density of RR intervals is decomposed into frequency bands:

$$\text{HF Power} = \int_{0.15}^{0.4} \text{PSD}(f)\, df \quad \text{(respiratory frequency band)} \tag{21}$$

$$\text{LF Power} = \int_{0.04}^{0.15} \text{PSD}(f)\, df \quad \text{(sympathetic/parasympathetic mix)} \tag{22}$$

$$\text{LF/HF ratio} = \frac{\text{LF Power}}{\text{HF Power}} \tag{23}$$

HF Power reflects parasympathetic (vagal) tone. High HF and low LF/HF indicate good emotional regulation and stress recovery capacity.

9. Social Bonding and Oxytocin-Vasopressin Coupling

9.1 Neurochemical Model of Attachment

Social bonding involves coordinated oxytocin (OXT) and vasopressin (AVP) signaling:

$$\frac{d[\text{OXT}]}{dt} = k_{\text{social}} \cdot \text{Social contact} - k_{\text{deg,OXT}} \cdot [\text{OXT}] \tag{24}$$

$$\frac{d[\text{AVP}]}{dt} = k_{\text{partner}} \cdot [\text{OXT}] - k_{\text{deg,AVP}} \cdot [\text{AVP}] \tag{25}$$

$$\text{Attachment strength} = \text{OXT}\cdot\text{AVP ratio} \times \text{Repeated pairing} \tag{26}$$

High OXT and AVP during positive social interaction (parent-infant, mating) strengthen pair bonds and reduce stress reactivity. Disrupted OXT/AVP signaling is implicated in social disorders (autism spectrum, social anxiety).

10. Case Studies and Simulations

10.1 Case Study 1: Multi-Dimensional Appraisal Scoring

Scenario: Two individuals react differently to the same job-loss event. Calculate their emotional responses using appraisal theory.

Given:

- Event: Unexpected job termination
- Individual A (resilient): $G_r = 0.7$, $G_c = -0.8$, $A_g = +0.3$ (partly internal), $C_p = 0.6$, $C_a = 0.4$
- Individual B (vulnerable): $G_r = 0.95$, $G_c = -0.95$, $A_g = -0.8$ (external blame), $C_p = 0.1$, $C_a = 0.0$
- Weights: $w_1 = 0.25, w_2 = 0.25, w_3 = 0.15, w_4 = 0.2, w_5 = 0.15$

Calculation for Individual A:

$$\begin{aligned} A_{\text{total}} &= 0.25(0.7) + 0.25(0.8) + 0.15(0.3) + 0.2(0.6) + 0.15(0.4) && (27)\\ &= 0.175 + 0.2 + 0.045 + 0.12 + 0.06 && (28)\\ &= 0.6 \text{ (moderate emotional intensity)} && (29) \end{aligned}$$

Intensity by formula: $I_A = 0.7 \times 0.8 \times (1 - 0.6) = 0.7 \times 0.8 \times 0.4 = 0.224$ (controlled response; can cope)

Calculation for Individual B:

$$\begin{aligned} A_{\text{total}} &= 0.25(0.95) + 0.25(0.95) + 0.15(0.8) + 0.2(0.1) + 0.15(0.0) && (30)\\ &= 0.2375 + 0.2375 + 0.12 + 0.02 + 0.0 && (31)\\ &= 0.6125 \text{ (higher emotional intensity)} && (32) \end{aligned}$$

Intensity: $I_B = 0.95 \times 0.95 \times (1 - 0.1) = 0.95 \times 0.95 \times 0.9 = 0.812$ (severe emotional response; cannot cope)

Emotional consequence:

Individual A appraises the job loss as significant but manageable → mixed (sadness, some anger) emotional response; emotion peaks briefly then resolves.

Individual B appraises it as catastrophic and uncontrollable → strong negative emotion (anxiety, despair); prolonged emotional reaction; risk of depression.

10.2 Case Study 2: Emotion Circumplex Mapping

Scenario: Map five emotions in valence-arousal space and calculate their intensities.

Given ($V_{\max} = 1.0, A_{\max} = 0.5, A_{\text{baseline}} = 0.3$):

Emotion	θ (°)	**Valence**	**Arousal**	**Intensity**
Joy	45	$\cos(45ř) = 0.707$	$0.3 + 0.5\sin(45ř) = 0.654$	$\sqrt{0.707^2 + 0.354^2} = 0.789$

Emotion	θ (°)	**Valence**	**Arousal**	**Intensity**
Sadness	225	$\cos(225\check{r}) = -0.707$	$0.3 + 0.5\sin(225\check{r}) = -0.054$	$\sqrt{0.5 + 0.003} = 0.709$
Fear	180	$\cos(180\check{r}) = -1.0$	$0.3 + 0.5\sin(180\check{r}) = 0.3$	$\sqrt{1 + 0.09} = 1.044$
Contentment	90	$\cos(90\check{r}) = 0$	$0.3 + 0.5\sin(90\check{r}) = 0.8$	$\sqrt{0 + 0.64} = 0.8$
Anger	135	$\cos(135\check{r}) = -0.707$	$0.3 + 0.5\sin(135\check{r}) = 0.654$	$\sqrt{0.5 + 0.427} = 0.965$

Interpretation: Fear and Anger have highest intensities (far from neutral point). Joy and Contentment are positive but vary in arousal. Sadness is low arousal negative emotion.

10.3 Case Study 3: Temporal Difference Learning Across Trial Blocks

Scenario: Model dopamine learning during a reward task (Tone predicts juice) across acquisition, overtraining, and extinction blocks. (See detailed case study calculations in document ending.)

Key findings:

- Acquisition ($\alpha = 0.3$): Value rises 0→0.97 in 10 trials; dopamine decreases 1.0→0.04
- Overtraining ($\alpha = 0.1$): Slower learning; dopamine remains low
- Extinction ($\alpha = 0.15$): Unlearning is slower than acquisition; dopamine goes negative (pause)

Asymmetry in learning rates explains why learned associations persist and why extinction therapy requires multiple trials.

10.4 Case Study 4: Fear Conditioning and Extinction Dynamics

Scenario: Model fear learning over 20 CS-US pairings, followed by 20 extinction trials.

Acquisition phase ($\beta_{\text{acq}} = 0.2$):

Trial	W **(synaptic strength)**	**Fear output**	W
1	0.0	0.0	0.2
2	0.2	0.2	0.16
3	0.36	0.36	0.128
5	0.59	0.59	0.082
10	0.878	0.878	0.024
20	0.98	0.98	0.004

After 20 pairings, $W \approx 0.98$ (nearly full learning); CS alone produces strong fear response.

Extinction phase ($\alpha_{\text{ext}} = 0.08$; slower than acquisition):

With extinction: Fear output = $W \times (1 - \text{Vmedia PFC}_{\text{inhibition}})$

Trial	**Vmedia PFC$_{\text{inhibition}}$**	**Fear output**	**Inhibition**
1	0.08	0.90	0.073
5	0.32	0.67	0.055
10	0.53	0.46	0.039
20	0.81	0.19	0.015

Interpretation: Extinction (20 trials) progresses more slowly than acquisition (20 trials reaches $W \approx 0.98$ but extinction only reduces fear to 0.19). Spontaneous recovery occurs because original W synapses remain potent; they are not erased, only masked by vmPFC inhibition.

10.5 Case Study 5: HPA Axis Cortisol Response to Stress

Scenario: Model cortisol time course during and after a 1-hour stress task.

Given:

- Baseline cortisol: 10 μg/dL
- Stress stimulus (0–60 min): $k_{\text{stim}} = 0.5$
- CRH degradation: $k_{\text{deg}} = 0.1$ min^{-1}
- Cortisol clearance: $k_{\text{clear}} = 0.05$ min^{-1}
- Negative feedback gain: $k_{\text{inhib}} = 0.1$

Stress phase (0–60 min):

| Time (min) | Stress | CRH (pg/mL) | ACTH (pmol/L) | Cortisol (μg/dL) | |—|—|—|—|—| | 0 | 1.0 | 5 | 20 | 10 | | 15 | 1.0 | 12 | 35 | 22 | | 30 | 1.0 | 16 | 42 | 35 | | 45 | 1.0 | 18 | 45 | 42 | | 60 (end stress) | 0.0 | 15 | 40 | 45 (peak) |

Recovery phase (60–180 min):

| Time (min) | Stress | CRH | ACTH | Cortisol | |—|—|—|—|—| | 90 | 0.0 | 8 | 25 | 35 | | 120 | 0.0 | 4 | 15 | 22 | | 180 | 0.0 | 2 | 10 | 12 (near baseline) |

Interpretation: Cortisol peaks 15 min after stress ends (delayed response due to HPA axis dynamics). Negative feedback suppresses CRH and ACTH, allowing cortisol to decline. Healthy stress recovery is complete within 2–3 hours. Chronic stress impairs negative feedback loop.

10.6 Case Study 6: Heart Rate Variability Analysis

Scenario: Analyze HRV from a 10-beat RR interval series in stressed vs. relaxed states.

Stressed state RR intervals (in milliseconds): 820, 800, 795, 810, 805, 792, 815, 798, 801, 808

Successive differences: 20, 5, 15, 5, 13, 23, 17, 3, 7 (total 9 differences)

$$\text{RMSSD} = \sqrt{\frac{1}{9}(400 + 25 + 225 + 25 + 169 + 529 + 289 + 9 + 49)} \tag{33}$$

$$= \sqrt{\frac{1721}{9}} = \sqrt{191.2} = 13.8 \text{ ms} \tag{34}$$

Interpretation: RMSSD = 13.8 ms indicates low vagal tone; sympathetic dominance during stress.

Relaxed state RR intervals: 850, 920, 875, 930, 900, 860, 945, 890, 910, 920

Successive differences: 70, 45, 55, 30, 40, 85, 55, 20, 10

$$\text{RMSSD} = \sqrt{\frac{1}{9}(4900 + 2025 + 3025 + 900 + 1600 + 7225 + 3025 + 400 + 100)} \tag{35}$$

$$= \sqrt{\frac{23200}{9}} = \sqrt{2577.8} = 50.8 \text{ ms} \tag{36}$$

Interpretation: RMSSD = 50.8 ms indicates high parasympathetic (vagal) tone; healthy emotional regulation and stress recovery capacity.

Therapeutic implication: HRV biofeedback training (learning to increase HRV) improves emotion regulation and reduces anxiety and depression symptoms.

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